DGMS Examination 2019 — Question 60 Detailed Solution

Junior Scientific Officer (Geology) | Plate Tectonics & Basin Analysis

QUESTION 60

A continental margin that forms due to initial rifting of a continent to develop an ocean later, is called:

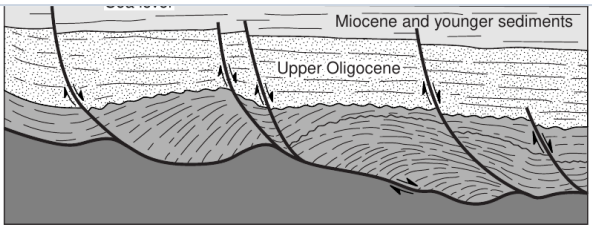
(a) Passive margin CORRECT ANSWER	(b) Active margin
(c) Sutured margin	(d) Andean-style margin

1. Executive Summary & Answer Verification

The correct option is **(a) Passive margin** (also known as an *Atlantic-type continental margin*).

In plate tectonics and sedimentary basin dynamics (the *Wilson Cycle*; McKenzie, 1978; Van der Pluijm & Marshak):

- When a continental landmass undergoes regional extension (continental rifting), crustal stretching and normal faulting thin the continental lithosphere.
- Continued extension eventually causes continental rupture (breakup). Decompression melting of asthenospheric mantle initiates seafloor spreading at a newly formed mid-ocean ridge, generating a young ocean basin (e.g., modern Red Sea).
- As seafloor spreading continues, the two rifted continental edges migrate away from the active spreading ridge. Because the boundary between the thinned continental crust and the new oceanic crust lies entirely *within the interior of the plate* (it is not a plate boundary), it experiences no subduction, no arc volcanism, and very little seismicity. Hence, it is universally termed a **Passive margin**.



(b)

FIGURE 16.28 Cross section of a passive-margin wedge. This cross section resembles that of the Gulf Coast (USA), but has been simplified, and is not an exact representation. Note that there are two levels of extensional faulting. The lower level formed during the original rifting of the continental crust that led to the formation of this passive margin. The upper level developed within the passive-margin wedge sediments and is due to seaward gravity-driven slip of the wedge. The detachment for this upper level probably lies in the salt near the base of the section. The salt has risen to form diapirs, especially in the footwalls of normal faults, due to buoyancy and to differential vertical loads. A thrust belt has developed at the toe of the wedge. The inset shows a detail of bedding in Cenozoic strata. Note that the faults are listric growth faults and that strata in the hanging-wall blocks thicken toward the fault.

cally, the weight of 1 km of sediment causes the floor of the basin to sink by about 0.33 km.

Stretching and thinning of continental lithosphere ceases after the rift–drift transition. Normal faults, however, do develop in a passive-margin sedimentary wedge, long after the drift stage has begun (Fig- is accommodated by normal faulting. Typically, the evaporite horizon near the base of the sedimentary section serves as the sliding surface (i.e., basal detachment) for this normal faulting (Figure 16.28).

Seaward sliding of passive-margin wedge sediment leads to the development of a system of half grabens,

Figure 60.1: *Architectural cross section of a **Passive Margin**: Stretched, rifted continental crust grades seaward into basaltic oceanic crust, buried beneath an enormous prograding sedimentary wedge (continental shelf, slope, and rise) formed by post-rift thermal subsidence. (Van der Pluijm & Marshak, Earth Structure, Figure 16.28, p. 407).*

2. Evolutionary Stages of a Passive Margin (The Rifting-to-Drifting Transition)

2.1 The McKenzie Stretching Model & Subsidence Phases

The formation of a passive margin follows two distinct, sequential geodynamic phases:

1. **Syn-Rift Phase (Mechanical Faulting):** Lithospheric stretching (extension factor $\beta > 1$) thins the crust by brittle planar or listric normal faulting in the upper crust and plastic flow in the lower crust. Fault-bounded rift grabens and half-grabens subside rapidly, accumulating coarse continental syn-rift clastics (alluvial fans, fluvial sands, lacustrine shales) and evaporites (e.g., Early Cretaceous pre-salt in the South Atlantic).
2. **Post-Rift Phase (Thermal Subsidence & Drifting):** Following continental breakup, the upwelled asthenosphere beneath the margin conducts heat away and cools exponentially. As the lithosphere cools, its density increases, causing regional isostatic sinking known as **thermal subsidence**:
   $$S(t) \propto \left(1 - \frac{1}{\beta}\right) \cdot \sqrt{t}$$
   This prolonged regional subsidence creates massive accommodation space, allowing rivers to deposit immense, seaward-thickening wedges of shallow-marine sands, carbonates, and deep-water turbidites (forming the continental shelf, continental slope, and continental rise).

2.2 Evaluation of the Other Margin Types

- **(b) Active Margin (Pacific-Type Margin):** Coincides directly with an active convergent plate boundary where oceanic lithosphere subducts beneath the continental margin (e.g., the western margin of South and North America). Characterized by deep oceanic trenches, accretionary wedges, forearc basins, high seismicity, and calc-alkaline volcanic arcs.

- **(c) Sutured Margin:** Represents the terminal stage of the Wilson Cycle, where an intervening ocean basin has completely closed and two continental landmasses have collided and fused together (e.g., the Indus-Yarlung Tsangpo Suture in the Himalayas, or the Uralian suture).
- **(d) Andean-Style Margin:** A specific archetype of an *active* convergent continental margin formed by oceanic-continental subduction, accompanied by a high-standing magmatic arc, retroarc fold-and-thrust belt, and deep seismic Wadati-Benioff zones.

3. Systematic Comparison of Continental Margins

Feature	Passive Margin (Atlantic-type)	Active Margin (Andean/Pacific-type)
Origin	**Continental rifting followed by ocean opening (Option a)**	Subduction of oceanic plate beneath continent
Relation to Plate Boundary	**Intraplate** (far from boundaries; no boundary at coastline)	**Plate boundary** (coincides with trench / subduction zone)
Tectonic & Seismic Activity	Tectonically quiescent; aseismic / rare intraplate quakes	Violent, high-magnitude megathrust earthquakes
Volcanism	Virtually absent (post-breakup)	Intense calc-alkaline arc volcanism (andesites, dacites)
Morphology	Broad continental shelf, gentle slope, wide rise	Narrow shelf, steep slope dropping into deep oceanic trench
Global Examples	East coast of North & South America, West coast of Africa, Indian east/west coasts	West coast of South America (Chile/Peru), Japan, Aleutians, Cascadia

Textbook Authority

Ben A. van der Pluijm & Stephen Marshak, *Earth Structure* (2nd ed., Chapter 16: Rifting, Seafloor Spreading, and Extensional Tectonics, p. 405–407):
"A passive margin is a continental margin that forms when continental rifting successfully separates a continent into two pieces and seafloor spreading creates a new ocean between them. Because the boundary between continental and oceanic crust is not a plate boundary, passive margins lack active plate-margin volcanism and seismicity."
Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy*:
"Passive continental margins develop during the initial rifting of a continent and the subsequent formation of an ocean basin... As the newly formed ocean basin widens, the continental margins cool and subside, accumulating thick prism-shaped wedges of sediment."